

DAMAGE TOLERANCE JUSTIFICATION OF COMPOSITE PARTS

Part: **4.9 – COMPOSITE PARTS**

Chapter: **4930**

Title: **DAMAGE TOLERANCE JUSTIFICATION OF COMPOSITE PARTS**

Revision: **0.1**

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1 INTRODUCTION

Certification Specifications for larger aircrafts (Ref. 2) propose a classification of the possible damage types in composite parts into five categories, each associated to a different level of requirements to complete demonstration of compliance.

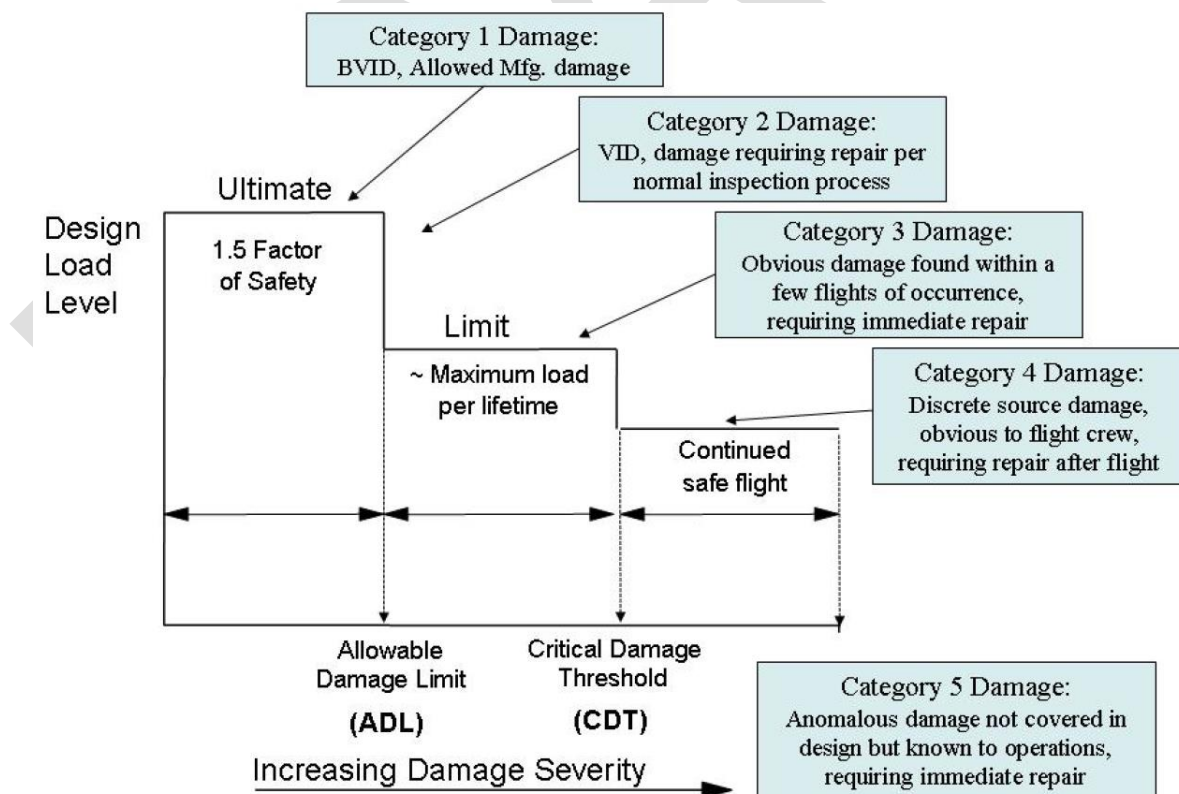


Figure 1. Schematic diagram of design load levels versus categories of damage severity

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Damage threats assessment performed for the composite part, as discussed in chapter 4920 "COMPOSITE DAMAGE THREATS", shall conclude with a set of damage types and severities which must be allocated within these categories proposed by regulators. As previously mentioned, different levels of justifications shall need to be addressed for damages within each category...

2 STRUCTURAL JUSTIFICATION AND DAMAGE TYPES

Within following sub-sections, the quotation marks enclose literal extracts of AMC 20-29 §8(a.)(1)(c).

2.1 DAMAGES OF CATEGORY 1

"Allowable damage that may go undetected by scheduled or directed field inspection and allowable manufacturing defects."

"Some examples of Category 1 damage include BVID and allowable defects caused in manufacturing or service (e.g., small delamination, porosity, small scratches, gouges, and minor environmental damage) that have substantiation data showing ultimate load is retained for the life of an aircraft structure."

Structural justification by analysis, considering the structure affected by damage so this kind, shall provide demonstration on...

- Ultimate load capability (for its lifetime, thus also...)
- No-damage growth along the whole service life

Structural analysis shall address this by:

- Material design values accounting for BVID damages in static strength demonstration up to UL (*) (and eventually limiting material stress/strain under no-damage-growth known limit under fatigue cases)
- Numerical FEM simulations accounting for BVID disbonds of bonded parts (e.g. stringers in stiffened panels) in static strength demonstration up to UL

Tests specimens (those on top of the tests pyramid) shall include a representative sample of the different possible types of the potentially critical defects/damages of this kind that may go undetected by inspections.

(*) Strength of bolted composite joints damaged by low velocity impacts

No comprehensive analytical methodology is really available to perform the strength analysis of bolted joints in composite laminates taking into account the effect of BVID undetected damages. The proposed strategy to deal with this topic in stress analysis shall be one of the following:

a) Missing fastener approach

For the most critical fastener location found in analysis using pristine material properties (that with minimum RF in the composite material), the fastener is assumed lost/missing in a second analysis. The residual strength of the remaining fastened attachment shall have to retain UL capability.

This approach is deemed to be significantly (overly) conservative.

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b) Penalty factors for design values

Design values (or reserve factors) calculated for the fastened joint in the composite material shall be affected by the following knock down factors...

✓ Net section failure

Both the bolt hole and the undetectable delamination damage can be considered "notches" (for sure of different nature and size). It is recognized in technical literature that a single notch uses to represent a higher degree of stress concentrations than multiple notches. Indeed it was observed that K_t 's (referenced with the net-section stress) decay with multiple damages and proximity (see Ref. 4). It is remarked that the size of the notches in this problem are typically very dissimilar (BVID much greater than fastener diameter) and the reduction of net section of the combined case compared with that with only the BVID will not be significant.

Thus, no further corrections shall be applied to the conventional analysis performed on the bolted joint, if in addition it is checked that the gross strain levels are below applicable TAI or CAI allowables.

✓ Bearing and Pull-Out

It is reasonable to expect that delaminations affecting the joint area will affect the strength of the laminate under bearing (shear in the bolt) and pull-out (tension in the bolt) stresses, as these failure modes are driven (up to some extent) by the matrix strengths within the laminate. On the other hand, bolt pretension will assure that plies under bolt-head/nut are in tight contact. It is also worth to mention that it is typical of standard test methods for composite bolted joints that fasteners installed in specimens are threaded bolts and that they are installed with just finger-tight torques.

The following correction factor is proposed for correcting the bearing and pull-out/through allowables of the pristine material) and shall conservatively cover BVID damages in a wide range of configurations (*):

$$KDF = 0.75$$

(*) Calculated assuming the "worst condition" of delaminations damage size up to 40mm (which should cover the BVID at any thickness, for either DVI & GVI detectability thresholds and a cut-off impact energy of 35J) and fastener nominal diameters from 4.8mm up to 7.9mm (as per Ref. 5, see §2.2.3 and §2.4).

2.2 DAMAGES OF CATEGORY 2

"Damage that can be reliably detected by scheduled or directed field inspections performed at specified intervals."

"Some examples of Category 2 damage include visible impact damage (VID), VID (ranging in size from small to large), deep gouges or scratches, manufacturing mistakes not evident in the factory, detectable delamination or debonding, and major local heat or environmental degradation that will sustain sufficient residual strength until found. This type of damage should not grow or, if slow or

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arrested growth occurs, the level of residual strength retained for the inspection interval is sufficiently above limit load capability.”

Representative damages of this category, in composite stiffened panels, can be:

- Impact damages at VID threshold
- Disbonds (an initial size of 100mm length is suggested for stiffeners bondings)

Structural justification by analysis, considering the structure affected by damage so this kind, shall provide demonstration on...

- No-damage growth or arrested growth between inspection intervals
- Limit load capability after the inspection interval, with the damage limited by arresting features
 - Only cases up to V_c to be considered
 - Material allowable values used in this justification shall account for BVID

Tests specimens (those on top of the tests pyramid), during the damage-tolerance demonstration phase, shall contain a minimum but sufficiently representative sample of the most critical damages of this kind.

2.3 DAMAGES OF CATEGORY 3

“Damage that can be reliably detected within a few flights of occurrence by operations or ramp maintenance personnel without special skills in composite inspection. Such damage must be in a location such that it is obvious by clearly visible evidence or cause other indications of potential damage that becomes obvious in a short time interval because of loss of the part form, fit or function. Both indications of significant damage warrant an expanded inspection to identify the full extent of damage to the part and surrounding structural areas. In practice, structural design features may be needed to provide sufficient large damage capability to ensure limit or near limit load is maintained with easily detectable, Category 3 damage”

“Some examples of Category 3 damage include large VID or other obvious damage that will be caught during walk-around inspection or during the normal course of operations (e.g., fuel leaks, system malfunctions or cabin noise).”

Representative damages of this category, in composite stiffened panels, can be:

- Full penetrating crack in skin between two adjacent stringers
- Stringer section broken

Structural justification by analysis, considering the structure affected by damage so this kind, shall provide demonstration on...

- Limit load capability
 - Only cases up to V_c to be considered
 - Material allowable values used in this justification shall account for BVID

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2.4 DAMAGES OF CATEGORY 4

“Discrete source damage from a known incident such as flight manoeuvres is limited”

“Some examples of Category 4 damage include rotor burst, bird strikes (as specified in the regulations), tyre bursts, and severe in-flight hail.”

Representative damages of this category in composite tail planes are:

- Full penetrating transverse crack affecting one stringer section and adjacent skin bays
- Full penetrating longitudinal crack affecting one rib/frame section and forward and rearward skin bays

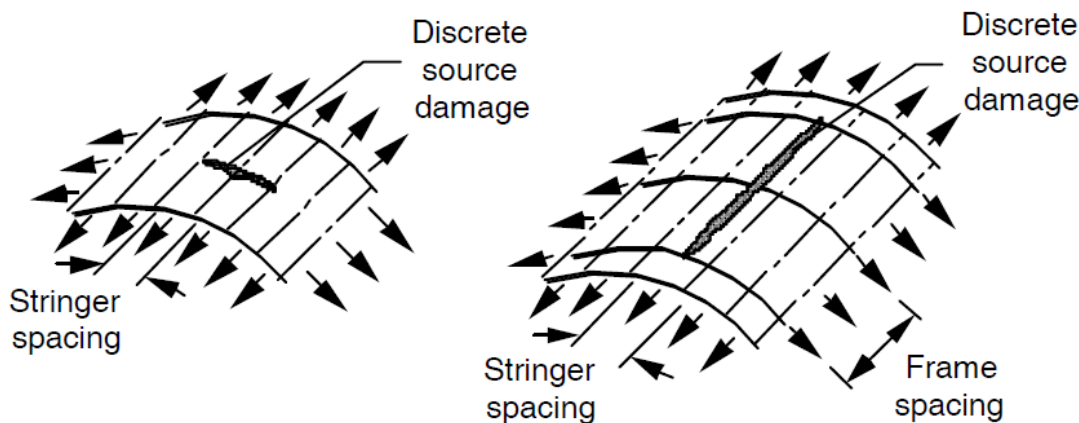


Figure 2. Schematics of discrete source damage (CMH-17 Vol. 3 §12.7.4.1)

- Bird strike (leading edge lost), severe in-flight hail (leading edge, or sandwich control surface, lost), uncontained fan blade impact, engine/rotating-machinery failure, fire, tire burst, lightning, ...

Structural justification by analysis, considering the structure affected by damage so this kind, shall provide demonstration on residual strength under get-home-loads (continued safe flight and landing, e.g. 70% of Limit for flight manoeuvre loads, see CMH-17, Vol. 3, §12.2.2.4). Material allowable values used in this justification shall account for BVID.

2.5 DAMAGES OF CATEGORY 5

“Some examples of Category 5 damage include severe service vehicle collisions with aircraft, anomalous flight overload conditions, abnormally hard landings, maintenance jacking errors, and loss of aircraft parts in flight, including possible subsequent high-energy, wide-area (blunt) impact with adjacent structure. Some Category 5 damage scenarios will not have clearly visual indications of damage, particularly in composite structures. However, there should be knowledge of other evidence from the related events that ensure safety is protected, starting with a complete report of possible damage by operations.”

No dedicated strength substantiation is required for damages of this kind.

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3 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
A/C	-	Aircraft (style "List of abbreviations")
BVID	-	Barely Visible Impact Damage. It is the damage in a composite laminate consequence of a low velocity impact which may occur at the threshold of (visible) detectability (limit of "Category 1" damages)
DET/DVI	-	Detailed Visual Inspection. Intensive visual examination requiring access, proximity, adequate lighting (grazing light) and any necessary inspection aids (such as mirror or hand lens)
DSG	-	Design Service Goal
DV	-	Design Value, i.e. the value for a mechanical property of a structural material to be used in stress analyses
FRP	-	Fibre Reinforced Polymer
GVI	-	General Visual Inspection, which may require some access (removal of fairings, access panels) and use of ladders or workstand to gain proximity
II	-	Inspection Interval
LL	-	Limit Load
HTP	-	Horizontal Tail Plane
NDT	-	Non Destructive Technique
VID	-	Visible Impact Damage. Threshold defining the maximum damage that needs to be considered as a "Category 2" damage.
VTP	-	Vertical Tail Plane
UL	-	Ultimate Load

REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	[EASA] CS-25	Amd.21	Certification Specifications and Acceptable Means of Compliance for Large Aeroplanes
Ref. 2	[EASA] AMC 20-29	07/2010	Composite Aircraft Structure (or FAA Advisory Circular AC No: 20-107B)
Ref. 3	[SAE] CMH-17	07/12	COMPOSITE MATERIALS HANDBOOK
Ref. 4	[R.E Peterson]	3rd Ed.	Stress concentration factors
Ref. 5	[Airbus] X029RP2011186	Is. 1.0	MANUFACTURING DEFECT KNOCK DOWN FACTOR FOR COMPOSITE PARTS

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SOFTWARE

Program	Version	Description

Table 1. Software programs referenced in this chapter

RECORD OF REVISIONS

Revision Date	Authors	Change Reason	Pages/Sub-chapters affected
0.1 23/02/2021	G. Baños	First issue	All pages

Table 2. Record of Revisions: authors, change reasons and sections/pages affected

APPROVAL SIGNATURES TABLE

Dpt. \ Site	Getafe	Manching
Stress Analysis (TEPAA)	Javier Gómez-Escalonilla DD/03/2021	Holger Hickethier DD/03/2021
Stress Methods (TEPAP-TL4)	María del Carmen Blanquer / Darío R. Fernández DD/03/2021	F. Glock DD/03/2021

Table 3. Approval signatures table for last revision of this chapter